

The Krebs Cycle

also the *citric acid cycle* · the *tricarboxylic acid (TCA) cycle* — the hub where every fuel the body burns finally converges

DEFINITION

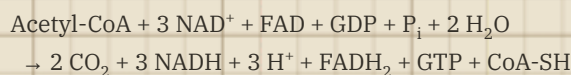
A closed loop of **eight enzyme-catalysed reactions** that completely oxidises the acetyl group of acetyl-CoA to **two molecules of CO₂**, capturing the released electrons as **NADH** and **FADH₂** and one high-energy phosphate as **GTP**.

The loop rebuilds its own starting material, oxaloacetate, so it acts catalytically and is never consumed. One oxaloacetate can carry an unlimited number of acetyl groups to CO₂.

KEY FACTS

- **Where** mitochondrial matrix — except succinate dehydrogenase, which is embedded in the inner membrane
- **Turns per glucose** 2
- **8 steps** · 4 oxidations · 2 decarboxylations · 1 substrate-level phosphorylation
- **Regulated at** steps 1, 3 and 4 — the three irreversible ones
- **Described by** Hans Krebs, 1937 · Nobel Prize 1953

NET EQUATION — ONE TURN



WHY THREE NAMES

Krebs cycle honours its discoverer. **Citric acid cycle** names the first product. **Tricarboxylic acid cycle** describes citrate's three -COO⁻ groups. All three mean the same pathway.

COMMON MISREADING

*The cycle **never touches O₂**. Oxygen is consumed downstream at Complex IV. The cycle is **aerobic by dependence, not by chemistry** — without the chain to regenerate NAD⁺ and FAD it stalls within seconds.*

WHY IT MATTERS

Carbohydrate, fat and protein all converge on acetyl-CoA. Whatever you eat, this is where its carbon is finally oxidised and its electrons harvested.

LEDGER FOR ONE TURN

3

NADH

1

FADH₂

1

GTP

2

CO₂

8

e⁻ OUT

4

OXIDATIONS

10

ATP EQUIV

20

PER GLUCOSE

ALL ROADS LEAD HERE

- **Carbohydrate** → pyruvate → acetyl-CoA
- **Fat** → β-oxidation → acetyl-CoA
- **Protein** → carbon skeletons → acetyl-CoA, α-KG, succinyl-CoA, fumarate, OAA
- **Ketone bodies** → acetyl-CoA

ORIENTATION · THE THREE STAGES

- **1 · Glycolysis** cytosol — glucose to pyruvate
 - **2 · Link + Krebs** matrix — carbon fully oxidised, electrons captured
 - **3 · Chain** inner membrane — electrons cashed into ATP
- ↳ *The cycle is stage 2 of 3. It harvests; it does not mint.*

EXAM POINT

The cycle turns **twice** per glucose because glycolysis yields two pyruvate. Every per-turn figure must be doubled before it can be quoted per glucose.

$3 \text{NADH} \cdot 1 \text{FADH}_2 \cdot 1 \text{GTP per turn} \Rightarrow 6 \cdot 2 \cdot 2 \text{ per glucose.}$

COMPARTMENTATION · WHAT HAS TO CROSS THE INNER MEMBRANE

- **Pyruvate** in, on the mitochondrial pyruvate carrier
- **Citrate** out, on the tricarboxylate carrier, to supply cytosolic acetyl-CoA
- **Malate / aspartate** both ways, to move reducing equivalents inward
- **Acetyl-CoA cannot cross** — it is made and consumed inside, or exported disguised as citrate
- **NADH cannot cross** — hence the shuttles, and hence 30 vs 32 ATP per glucose
- **CoA and NAD⁺ pools are separate** on the two sides of the membrane

KEY TAKEAWAY

*The Krebs cycle is not primarily an ATP factory — it makes just one nucleotide triphosphate per turn. It is an **electron harvester** and a **biosynthetic roundabout**; its true output is reduced coenzyme for the respiratory chain, and its intermediates are the raw material for fats, haem, amino acids and glucose.*

The wheel



Eight steps, four oxidations, two carbons in and two carbons out. Enzyme names sit outside the ring; what each step releases is printed beneath them in ochre.

READING THE WHEEL

- **Teal ring** the carbon skeleton, always 4 → 6 → 5 → 4
- **Ochre labels** energy leaving as reduced coenzyme
- **Two CO₂** leave at steps 3 and 4 only

ORDER MNEMONIC

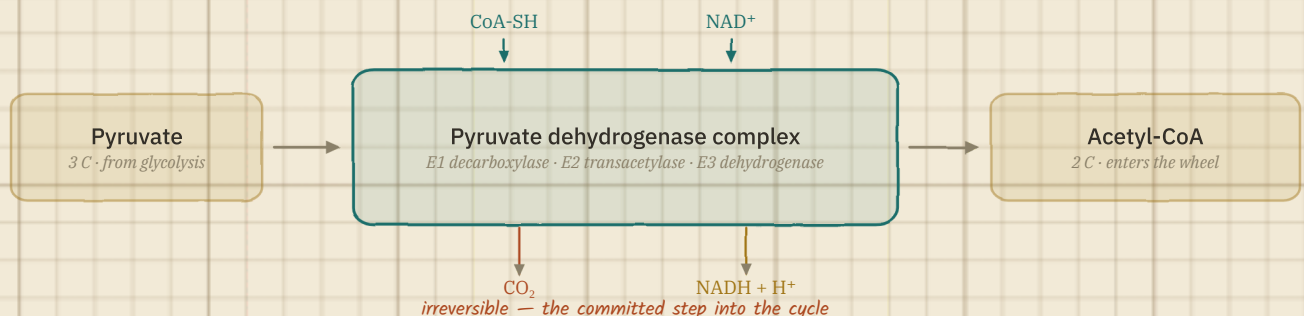
Citrate Is Krebs' Starting Substrate For Making Oxaloacetate

Citrate · Isocitrate · α-Ketoglutarate · Succinyl-CoA · Succinate · Fumarate · Malate · Oxaloacetate

WHERE THE ENERGY GOES

3 NADH × 2.5 + 1 FADH₂ × 1.5 + 1 GTP = **10 ATP per turn**, so **20 ATP per glucose** come from the cycle alone.

Step zero — the link reaction



THREE ENZYMES, ONE MACHINE

- **E1** pyruvate dehydrogenase — decarboxylates, needs TPP
- **E2** transacetylase — swings the acetyl group on a lipoamide arm
- **E3** dehydrogenase — re-oxidises lipoamide, hands electrons to NAD⁺

FIVE COFACTORS, ALL NEEDED

Thiamine pyrophosphate · Lipoamide · CoA-SH · FAD · NAD⁺

"Tender Loving Care For Nancy"

↳ Three are vitamin-derived — why deficiency hits this step first.

WHY IT IS THE POINT OF NO RETURN

Once pyruvate is decarboxylated its carbons can never return to glucose. This single irreversible step is the reason **fatty acids cannot be converted to glucose** in humans.

The eight steps

#	REACTION	ENZYME	CHEMISTRY	YIELD	ΔG°
1	Acetyl-CoA + oxaloacetate + H ₂ O → citrate + CoA-SH	citrate synthase	Claisen condensation; hydrolysis of the thioester bond supplies the driving force	—	-32.2
2	Citrate ⇌ <i>cis</i> -aconitate ⇌ isocitrate	aconitase	Dehydration then rehydration on the opposite face; Fe-S cluster. Moves the -OH onto a carbon that <i>can</i> be oxidised	—	+13.3
3	Isocitrate + NAD ⁺ → α-ketoglutarate + CO ₂	isocitrate dehydrogenase	Oxidative decarboxylation via an oxalosuccinate intermediate — rate-limiting	NADH · CO₂	-20.9
4	α-Ketoglutarate + NAD ⁺ + CoA-SH → succinyl-CoA + CO ₂	α-ketoglutarate dehydrogenase complex	Oxidative decarboxylation; mechanistically a twin of the PDH complex — the same five cofactors	NADH · CO₂	-33.5
5	Succinyl-CoA + GDP + P _i → succinate + GTP + CoA-SH	succinyl-CoA synthetase	The cycle's only substrate-level phosphorylation — thioester energy captured directly as a nucleotide triphosphate	GTP	-2.9
6	Succinate + FAD → fumarate + FADH ₂	succinate dehydrogenase (Complex II)	FAD is used because the C-C dehydrogenation is not energetic enough to reduce NAD ⁺ ; electrons pass straight to ubiquinone	FADH₂	≈ 0
7	Fumarate + H ₂ O → L-malate	fumarase	Stereospecific <i>trans</i> hydration — produces only the L isomer, never D	—	-3.8
8	L-Malate + NAD ⁺ → oxaloacetate + NADH + H ⁺	malate dehydrogenase	Alcohol oxidised to a ketone, regenerating the acceptor and closing the loop	NADH	+29.7

ΔG° in kJ·mol⁻¹ at standard conditions (Lehninger values; texts vary by a few kJ). Steps **1, 3 and 4** are strongly exergonic and effectively irreversible — they are the cycle's control points.

WHY STEP 8 RUNS UPHILL

Malate dehydrogenase has $\Delta G^{\circ} = +29.7 \text{ kJ} \cdot \text{mol}^{-1}$ — thermodynamically it should run backwards. It does not, because citrate synthase consumes oxaloacetate the instant it appears, holding matrix [OAA] below **~1 μM**.

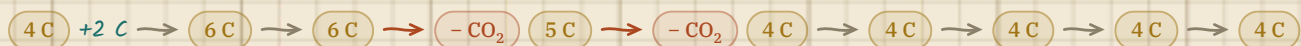
The cycle is pulled forward by product removal, not pushed by each step. Coupling, not brute force.

FOUR KINDS OF CHEMISTRY, THAT IS ALL

- **Condensation** step 1 — builds the six-carbon skeleton
- **Isomerisation** step 2 — relocates one hydroxyl
- **Oxidative decarboxylation** steps 3, 4 — the two CO₂ and two of the NADH
- **Oxidation / hydration** steps 6–8 — rebuild oxaloacetate and take the last NADH

↳ Step 5 is the outlier: the one place the cycle makes a phosphate bond directly.

Following the carbons



THE SUBTLETY EXAMINERS LOVE

The two CO₂ released in a given turn are **not** the two carbons that just arrived as acetyl-CoA. Isotope labelling shows both released carbons originate from the **oxaloacetate** already in the cycle: C4 of oxaloacetate becomes C1 of α-ketoglutarate and leaves at step 4. The acetyl carbons are retained in the four-carbon skeleton and leave on *later* turns.

↳ Carbon bookkeeping balances every turn (2 in, 2 out) — but the individual atoms do not.

WHERE EACH PRODUCT APPEARS

- **NADH** steps 3, 4, 8
- **FADH₂** step 6 only
- **GTP** step 5 only
- **CO₂** steps 3, 4 only
- **H₂O consumed** steps 1, 7

COFACTOR CHECK

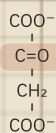
Only two steps need anything exotic: aconitase carries an **Fe-S cluster**, and α-KG dehydrogenase needs the same **five cofactors** as PDH. Everything else runs on NAD⁺, FAD, CoA and water.

Nine molecules

Every intermediate is a short carbon chain capped with carboxylates. Only one group changes at each step — follow the highlight and the cycle becomes a single running edit to one molecule.

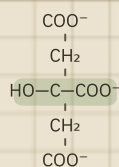
hydroxyl carbonyl thioester
double bond

1 Oxaloacetate 4 C



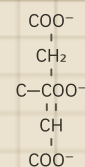
KETONE
regenerated every turn

2 Citrate 6 C



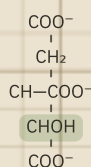
3° ALCOHOL
tertiary · not oxidisable

3 cis-Aconitate 6 C



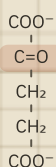
ENZYME-BOUND
bound intermediate

4 Isocitrate 6 C



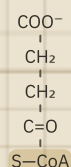
2° ALCOHOL
secondary · now oxidisable

5 α-Ketoglutarate 5 C



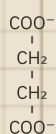
A-KETO ACID
first CO_2 has left

6 Succinyl-CoA 4 C



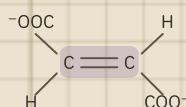
THIOESTER
the energy store

7 Succinate 4 C



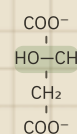
SYMMETRIC
no chiral centre left

8 Fumarate 4 C



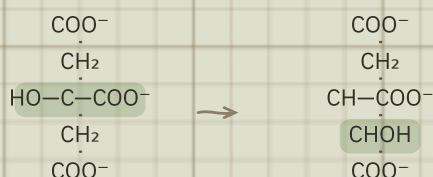
trans (E)
ALKENE
trans only · not maleate

9 L-Malate 4 C



L ONLY
water added across $\text{C}=\text{C}$

WHY STEP 2 EXISTS — AN ENTIRE ENZYME TO MOVE ONE HYDROXYL



Citrate is a dead end: its hydroxyl sits on a carbon carrying **no hydrogen**, and an alcohol cannot be oxidised to a carbonyl without a hydrogen to remove. Aconitase dehydrates to *cis*-aconitate and rehydrates the other way round, moving the $-\text{OH}$ to the neighbouring carbon. Isocitrate is a **secondary** alcohol — and step 3 can proceed.

OGSTON'S PUZZLE · 1948

Citrate is achiral and looks symmetric — two identical $-\text{CH}_2\text{COO}^-$ arms. Yet labelling shows the cycle always attacks *the same arm*, the one from oxaloacetate. Ogston resolved it: an enzyme binding a **prochiral** substrate at **three points** tells two identical-looking groups apart, because only one orientation fits.

WHY THIOESTERS

An oxygen ester is stabilised by resonance. Sulfur's 3p orbitals overlap poorly with carbon's 2p, so a **thioester gets almost none of that stabilisation** and hydrolyses with $\Delta G^\circ \approx -31 \text{ kJ}\cdot\text{mol}^{-1}$ — on a par with ATP.

↳ Cashed twice: at step 1, and again at step 5.

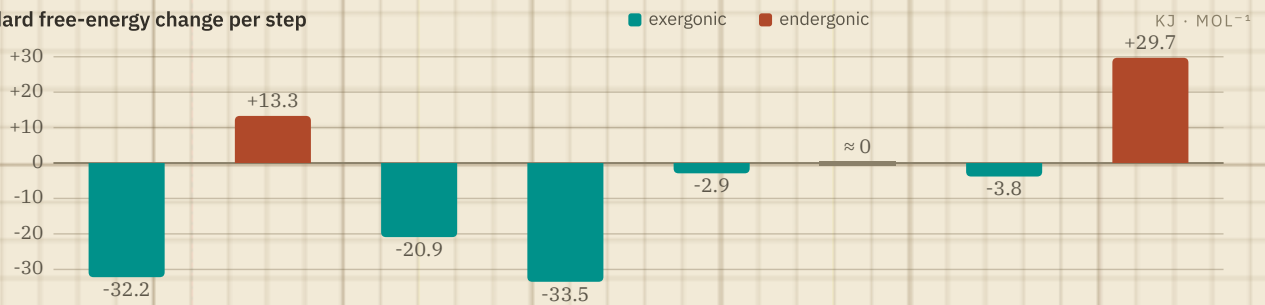
STEREOSPECIFICITY

Fumarase adds water *anti* across the double bond and accepts only the *trans* isomer, so the product is exclusively **L-malate** — maleate is not a substrate.

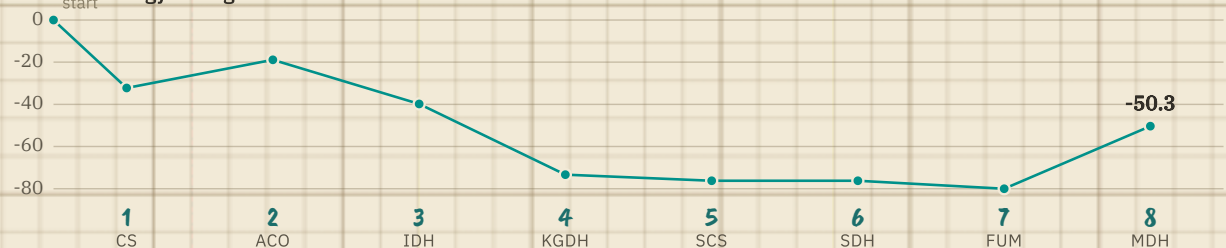
↳ Succinate is **symmetric** — where the acetyl carbons lose their identity.

The energy landscape

Standard free-energy change per step



Cumulative free energy through one turn



Three steps do nearly all the thermodynamic work. Step 8 climbs 29.7 kJ · mol⁻¹ back up and still runs forward, because citrate synthase keeps oxaloacetate scarce.

WHAT ΔG° DOES NOT TELL YOU

Standard values assume 1 M of everything at pH 7 — conditions no cell meets. In a working mitochondrion only steps **1, 3 and 4** stay far from equilibrium; the other five sit near $\Delta G \approx 0$.

A reaction at equilibrium cannot control flux — only a displaced one can.

NET FOR ONE TURN

$$\Sigma \Delta G^\circ = -50.3 \text{ kJ} \cdot \text{mol}^{-1}$$

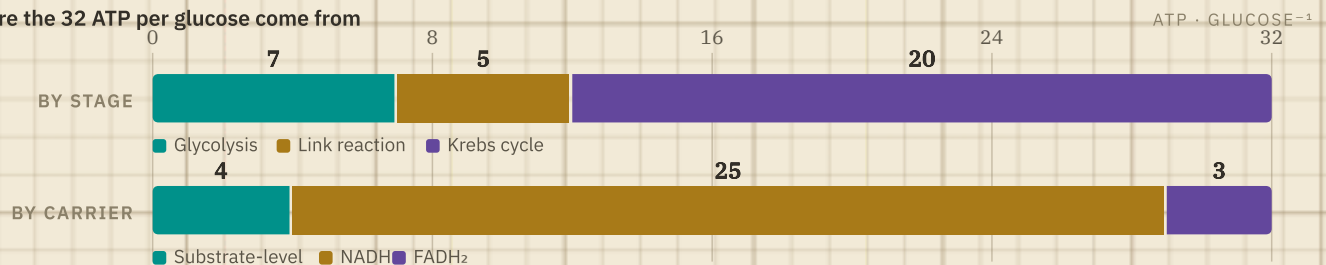
Barely one ATP's worth of free energy. Almost everything useful leaves as **reduced coenzyme** — the cycle is a harvester, not a mint.

READING THE LOWER PANEL

The running total falls steeply through steps 1–4, flattens across 5–7, then climbs at step 8 — the cycle descends into an energy well and pays a little back to close the loop.

Where the ATP comes from

Where the 32 ATP per glucose come from



Liver and heart, using the malate–aspartate shuttle. The same glucose yields 30 ATP in brain and skeletal muscle.

WORKED EXAMPLE · ATP PER GLUCOSE

GIVEN	1 glucose, liver, malate–aspartate shuttle. P/O 2.5 per NADH, 1.5 per FADH ₂
FORMULA	$\text{ATP} = \text{ATP}_{\text{SLP}} + 2.5 \cdot n(\text{NADH}) + 1.5 \cdot n(\text{FADH}_2)$
SUBSTITUTE	Glycolysis 2 ATP + 2 NADH · Link ×2 → 2 NADH · Krebs ×2 → 6 NADH + 2 FADH ₂ + 2 GTP ⇒ SLP 4 · NADH 10 · FADH₂ 2
CALCULATE	$4 + (10 \times 2.5) + (2 \times 1.5) = 4 + 25 + 3$
RESULT	32 ATP per glucose
INTERPRET	The cycle supplies 20 of the 32 yet mints only 2 directly. 30 in brain and muscle, where the glycerol-3-phosphate shuttle delivers cytosolic NADH as FADH ₂ .

READ BOTH BARS TOGETHER

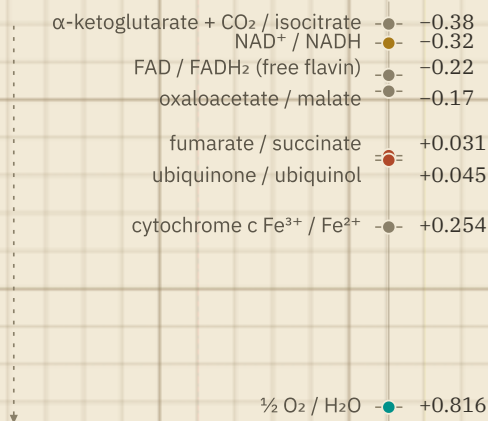
The cycle contributes **20 of the 32** — but on the second bar, **25 of the 32** arrive as NADH and only **4** by direct phosphorylation.

WHY THE NUMBERS ARE NOT INTEGERS

Older texts said 3 ATP per NADH. Proton counting gives **≈ 10 H⁺ pumped per NADH** and **4 H⁺ per ATP** including transport, hence **2.5** — a ratio, not a stoichiometry.

Why FADH₂ is worth less

Standard reduction potential E° / volts



NADH → O₂

$$\Delta E^\circ = 1.14 \text{ V} \cdot \Delta G^\circ = -219 \text{ kJ} \cdot \text{mol}^{-1}$$

≈ **2.5 ATP**

succinate → O₂

$$\Delta E^\circ = 0.79 \text{ V} \cdot \Delta G^\circ = -152 \text{ kJ} \cdot \text{mol}^{-1}$$

≈ **1.5 ATP**

Electrons only fall from a more negative couple to a more positive one. The size of the fall sets the ATP yield.

THE ANSWER EVERYONE ASKS FOR

Fumarate/succinate sits at **+0.031 V** — about **0.35 V more positive** than NAD⁺/NADH. Succinate **cannot** reduce NAD⁺: the electrons would have to run uphill.

So succinate dehydrogenase uses a tightly bound **FAD** and passes its electrons to ubiquinone at +0.045 V, entering the chain *past* Complex I. One proton-pumping site is skipped and the yield falls from ≈ 2.5 ATP to ≈ 1.5.

THE FORMULA BEHIND THE LADDER

$$\Delta G^\circ = -nF\Delta E^\circ$$

$$n = 2 \text{ electrons} \cdot F = 96.5 \text{ kJ} \cdot \text{V}^{-1} \cdot \text{mol}^{-1}$$

A 1 V drop across two electrons releases ≈ 193 kJ·mol⁻¹. Every yield figure on these sheets follows from that one relation — the potentials are measured, the ATP counts are derived.

Getting cytosolic electrons in

Glycolysis makes NADH in the cytosol, but the inner membrane is impermeable to it. Two shuttles move the *reducing equivalents* instead of the molecule — and which one a tissue uses decides whether glucose is worth 30 ATP or 32.

MALATE-ASPARTATE SHUTTLE

- Cytosolic NADH reduces oxaloacetate to malate; malate crosses and is re-oxidised **back to NADH** inside
- Aspartate and α-ketoglutarate carry the carbon skeletons back out
- **Yield preserved** — 2.5 ATP per cytosolic NADH
- Used by **liver, kidney and heart**

GLYCEROL-3-PHOSPHATE SHUTTLE

- Cytosolic NADH reduces DHAP to glycerol-3-phosphate
- A membrane-bound flavoenzyme re-oxidises it, handing electrons to **FAD, then ubiquinone**
- **Yield lost** — 1.5 ATP per cytosolic NADH, because Complex I is bypassed
- Used by **brain and skeletal muscle**; fast, and irreversible

EVERY CARRIER THE CYCLE PRODUCES, AND WHAT IT IS WORTH

CARRIER	MADE AT	ENTERS THE CHAIN AT	COMPLEXES THAT PUMP	ATP
NADH	steps 3, 4, 8 (and the link reaction)	Complex I	I, III, IV	≈ 2.5
FADH₂	step 6, on succinate dehydrogenase	ubiquinone, via Complex II	III, IV	≈ 1.5
GTP	step 5, substrate-level	—	none	1

↳ Complex II is the only one of the four that pumps no protons — it is an entry point, not a pump. That single fact is the whole 2.5-versus-1.5 story.

PUTTING THE TWO PAGES TOGETHER

Every ATP figure quoted for glucose is really three separate claims: how many reduced carriers are made (fixed by the chemistry), how far their electrons fall (fixed by the potentials above), and how efficiently the proton gradient is converted (the P/O ratio). Change the shuttle and only the second claim moves — which is the whole difference between 30 and 32.

Three valves, one signal

Unlike glycolysis, the cycle has no single committed step. It is governed by **substrate supply and product inhibition** at the three irreversible reactions — and every input reduces to one question: *does the cell need energy right now?*

VALVE 1 · STEP 1

Citrate synthase

- ▼ ATP, NADH
- ▼ succinyl-CoA (product-like)
- ▼ citrate (own product)
- ▼ long-chain acyl-CoA

Mostly limited by how much acetyl-CoA and oxaloacetate are available.

VALVE 2 · STEP 3 · RATE-LIMITING

Isocitrate dehydrogenase

- ▲ ADP, Ca^{2+}
- ▼ ATP, NADH

The true throttle — allosterically sensitive to ADP, so it reads the energy charge directly.

VALVE 3 · STEP 4

α -KG dehydrogenase

- ▲ Ca^{2+}
- ▼ succinyl-CoA, NADH, ATP

Inhibited by its own product — the same logic as the PDH complex it resembles.

UPSTREAM GATE · COVALENT CONTROL

The PDH complex is switched, not throttled

PDH kinase phosphorylates and **switches it OFF** — activated by ATP, NADH and acetyl-CoA (plenty of fuel already). PDH phosphatase dephosphorylates and **switches it ON** — activated by Ca^{2+} and, in adipose tissue, insulin.

↳ *Allostery tunes the cycle; phosphorylation gates what enters it.*

THE RULE THAT COVERS ALL OF IT

High **NADH / NAD⁺** and high **ATP / ADP** $\Rightarrow$ the cycle slows.

Ca^{2+} — the signal that muscle is contracting — opens **three enzymes at once** (PDH, IDH, α -KGDH), so ATP supply rises exactly when demand does.

WHY REGULATION SITS WHERE IT DOES

- Steps **1, 3, 4** are far from equilibrium — displaced reactions can carry control
- Steps **2, 5, 6, 7, 8** sit near $\Delta G \approx 0$ — they follow, they do not lead
- Control is therefore **distributed**, not vested in one committed step as in glycolysis

CONTRAST · GLYCOLYSIS

Glycolysis has one dominant valve, phosphofructokinase-1, and a clear committed step. The cycle has none: it is paced by **how fast the respiratory chain removes NADH**. Regulation here is a consequence of demand, not a decision point.

↳ *Which is why “what is the rate-limiting step of the TCA cycle?” has a softer answer than the same question about glycolysis.*

SIGNAL SUMMARY

SIGNAL	MEANS	EFFECT ON THE CYCLE
↑ ADP	energy is being spent	activates IDH — speeds up
↑ Ca^{2+}	muscle is contracting	activates PDH, IDH, α -KGDH — speeds up
↑ ATP	energy is plentiful	inhibits CS and IDH — slows
↑ NADH	chain cannot keep up	inhibits CS, IDH, α -KGDH — slows
↑ succinyl-CoA	downstream backlog	inhibits CS and α -KGDH — slows

SCENARIO · WHAT HAPPENS IN THE FIRST SECONDS OF A SPRINT

- **0 s** — myosin ATPase consumes ATP; ADP and P_i rise in the matrix
- **Instantly** — Ca^{2+} released for contraction also floods the matrix
- **Ca^{2+} opens three enzymes** — PDH phosphatase switches PDH on; IDH and α -KGDH are activated directly
- **ADP relieves inhibition** of isocitrate dehydrogenase, and drives ATP synthase directly
- **NADH is consumed faster** by the chain, lifting product inhibition on all three valves
- **Net effect** — flux can rise many-fold within seconds, with no change in enzyme amount and no hormone involved

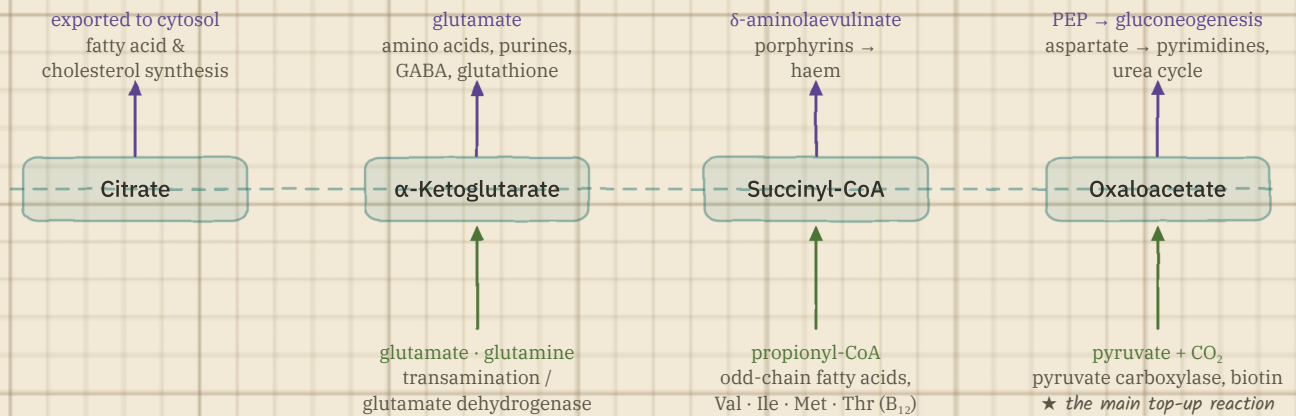
↳ *The same Ca^{2+} signal that orders the contraction also pays for it. Supply and demand share one messenger.*

PUTTING IT TOGETHER

The cycle has no master switch because it does not need one. Every regulator is either a **product** (NADH, succinyl-CoA, citrate) or a **proxy for demand** (ADP, Ca^{2+}). Supply therefore tracks demand automatically, and the respiratory chain — by consuming NADH — sets the pace of everything upstream of it. **Stop the chain and the cycle stops within seconds**, not because oxygen is a substrate, but because NAD^+ never comes back.

An amphibolic roundabout

Amphibolic = catabolic and anabolic at once. Intermediates are constantly withdrawn for biosynthesis (**cataplerosis**) and must be replaced (**anaplerosis**) or the wheel runs dry.



▲ violet — carbon leaves for biosynthesis · ▲ green — carbon is replaced

ANAPLEROTIC · FILLING THE CYCLE

REACTION	ENTERS AS	NOTE
Pyruvate carboxylase pyruvate + CO ₂ → OAA	oxaloacetate	Biotin; activated by acetyl-CoA. The main one
Glutamate dehydrogenase / transaminases	α-ketoglutarate	Links amino-acid nitrogen to the cycle
Propionyl-CoA pathway	succinyl-CoA	Odd-chain fats, Val/Ile/Met/Thr; needs B ₁₂
Aspartate transamination	oxaloacetate	Reversible; ties into the urea cycle

CATAPLEROTIC · DRAWING ON THE CYCLE

INTERMEDIATE	LEAVES TO MAKE	NOTE
Citrate	fatty acids, cholesterol	Citrate shuttle carries acetyl units to the cytosol
α-Ketoglutarate	glutamate, GABA, purines, glutathione	The nitrogen gateway
Succinyl-CoA	haem	Condenses with glycine to δ-aminolaevulinate
Oxaloacetate	glucose, aspartate, pyrimidines	Via PEP carboxykinase into gluconeogenesis

CLINICAL CONSEQUENCE · WHY FATS BURN IN THE FLAME OF CARBOHYDRATES

Acetyl-CoA **cannot** replenish oxaloacetate — it only consumes it. In starvation or uncontrolled diabetes, oxaloacetate is drained into gluconeogenesis; the cycle cannot keep pace with the acetyl-CoA arriving from fat breakdown, and the surplus is diverted into **ketone bodies**.

The same arithmetic explains why **fatty acids cannot be converted to glucose** in humans: both acetyl carbons are lost as CO₂ before oxaloacetate is regenerated, so there is no net gain to feed gluconeogenesis.

HOW ACETYL UNITS ACTUALLY GET OUT

Acetyl-CoA cannot cross the inner membrane, so it leaves **disguised as citrate**. In the cytosol, ATP-citrate lyase splits citrate back into acetyl-CoA and oxaloacetate; the oxaloacetate returns as malate or pyruvate. This **citrate shuttle** is how every fatty acid you synthesise gets its carbon.

BALANCE RULE

*Every intermediate withdrawn must be replaced. Withdraw without topping up and flux collapses — the cycle is **catalytic**, not **stoichiometric**.*

THREE WORDS TO BE ABLE TO DEFINE

- **Amphibolic** serving catabolism and anabolism simultaneously
- **Anaplerotic** a reaction that replenishes a cycle intermediate
- **Cataplerotic** withdrawal of an intermediate for biosynthesis

Where the wheel jams

AGENT	TARGET	CONSEQUENCE
Fluoroacetate (rodenticide 1080)	aconitase — step 2	Converted in the cell to fluorocitrate, which blocks the enzyme: “lethal synthesis”. Citrate accumulates; the cycle halts
Arsenite & trivalent arsenicals	lipoamide — PDH and α -KGDH	Binds the two neighbouring thiols of dihydrolipoamide, disabling both complexes at once
Malonate	succinate dehydrogenase — step 6	Classic competitive inhibitor and structural analogue of succinate; Krebs's own evidence that the pathway is a cycle
3-Nitropropionic acid	succinate dehydrogenase — irreversible	Selective striatal degeneration; used experimentally to model Huntington disease
Thiamine (B₁) deficiency	TPP-dependent enzymes — PDH, α -KGDH, transketolase	Pyruvate cannot enter the cycle → lactic acidosis. Wet and dry beriberi, Wernicke encephalopathy, Korsakoff syndrome

Neurons and cardiac myocytes fail first — highest oxidative demand, least glycolytic reserve.

ONCOMETABOLITES · CYCLE ENZYMES AS TUMOUR SUPPRESSORS

- **IDH1 / IDH2** — a neomorphic activity converts α -KG into **D-2-hydroxyglutarate**, blocking α -KG-dependent dioxygenases and driving hypermethylation. Gliomas, AML, chondrosarcoma
- **SDH (A–D)** — succinate accumulates; hereditary paraganglioma, pheochromocytoma, GIST, with pseudohypoxia via HIF-1 α
- **FH** — fumarate accumulates and succinates cysteines, aconitase included; hereditary leiomyomatosis and renal cell carcinoma

INHERITED DEFECTS & IMMUNOMETABOLISM

- **PDH deficiency** — congenital lactic acidosis; a ketogenic diet helps because ketones enter downstream of the block
- **MDH2 mutations** — early-onset encephalopathy, refractory epilepsy, raised lactate
- **Activated macrophages** deliberately break the cycle so succinate drives IL-1 β and citrate becomes anti-inflammatory **itaconate**

SIX THINGS PEOPLE GET WRONG

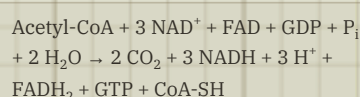
- **“The cycle uses oxygen.”** It does not. O₂ is the final acceptor at Complex IV; the cycle depends on that only to get NAD⁺ and FAD back
- **“NADH gives 3 ATP, FADH₂ gives 2.”** Old integer P/O ratios. Current values are **2.5** and **1.5**
- **“GTP counts as two ATP.”** One GTP $\equiv$ one ATP — nucleoside diphosphate kinase interconverts them freely
- **“The acetyl carbons leave as that turn's CO₂.”** They do not — the released carbons come from oxaloacetate
- **“Acetyl-CoA can be turned into glucose.”** Not in humans. Both carbons are lost as CO₂ before oxaloacetate is regenerated
- **“Succinate dehydrogenase sits in the matrix.”** It is the one cycle enzyme bound to the inner membrane — it is Complex II

ONE-GLANCE RECAP

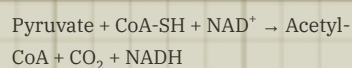
KEY NUMBERS

Steps · oxidations · decarboxylations	8 · 4 · 2
Per turn	3 NADH · 1 FADH ₂ · 1 GTP · 2 CO ₂
ATP per turn	10
Turns per glucose	2
ATP per glucose, from the cycle	20
ATP per glucose, total	30–32
Rate-limiting enzyme	isocitrate dehydrogenase

FORMULAS WORTH MEMORISING



$$\text{ATP} = \text{ATP}_{\text{SLP}} + 2.5 n(\text{NADH}) + 1.5 n(\text{FADH}_2)$$



THE THREE VALVES

- **Citrate synthase** — off with ATP, NADH, succinyl-CoA, citrate
- **Isocitrate dehydrogenase** — on with ADP and Ca²⁺, off with ATP and NADH
- **α -KG dehydrogenase** — on with Ca²⁺, off with succinyl-CoA, NADH, ATP

*Final takeaway — the cycle is an **electron harvester** and a **biosynthetic roundabout**, not an ATP factory.*

SOURCES CONSULTED

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